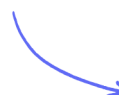


Structured Questions

The Periodic Table

Periodicity - Ionisation Energy / Periodicity - Thermal Trends

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Total Marks

/20

1 (a) Aluminium is an abundant metal with many uses.

The first four ionisation energies of aluminium are shown.

Ionisation number	Energy / kJ mol^{-1}
1	578
2	1820
3	2750
4	11600

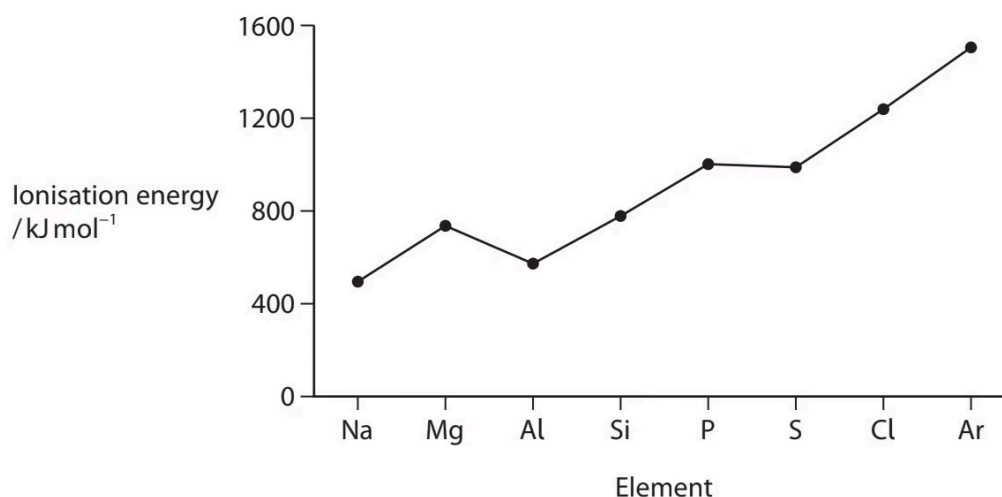
Explain how this information shows that aluminium is in Group 3 of the Periodic Table.

.....

.....

(2 marks)

(b) The graph shows the first ionisation energies for the elements in Period 3.



i) Explain the general increase in the first ionisation energy from sodium to argon.

(2)

ii) Explain why the first ionisation energy of aluminium is less than the first ionisation energy of magnesium.

(2)

(4 marks)

(c) i) Describe the bonding in aluminium metal.

(2)

ii) Give two possible reasons why aluminium is used for overhead power cables.

(2)

(4 marks)

(d) New uses for waste aluminium cans are being investigated. One possible use is to make nanoparticle alloys to produce hydrogen for fuel.

i) Aluminium nanoparticles react with water to produce aluminium oxide and hydrogen.

Complete the following equation. State symbols are not required.

(1)

..... Al + H₂O → +

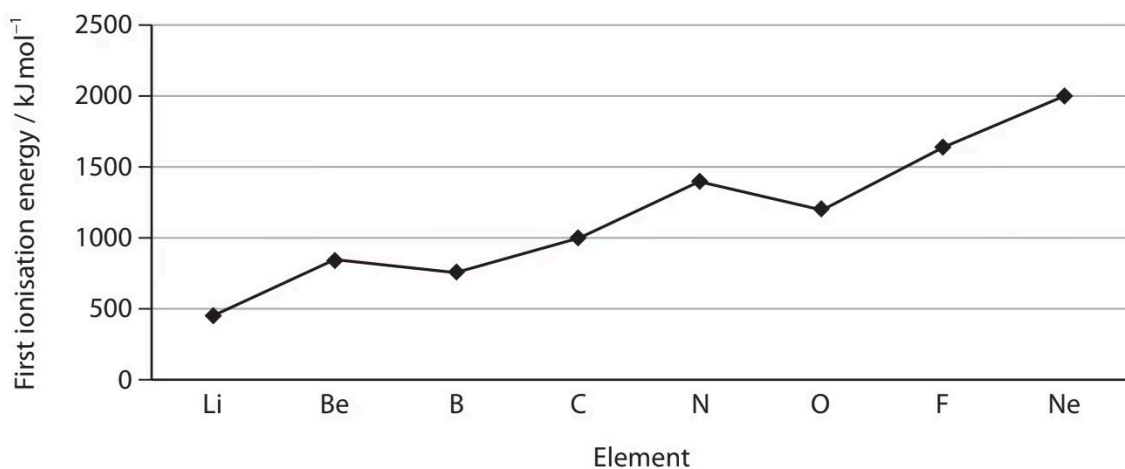
ii) Give **two** possible reasons for producing hydrogen from aluminium rather than from fossil fuels.

(2)

(3 marks)

2 (a) This question is about the ionisation energies of the elements in Period 2 of the Periodic Table.

The first ionisation energies of the Period 2 elements are shown.



i) Give an equation that represents the first ionisation energy of lithium.
Include state symbols. **(1)**

ii) Explain why there is a general increase in the first ionisation energy across the period.

(2)

iii) Explain why the first ionisation energy of oxygen is lower than that of nitrogen.

(2)

(5 marks)

(b) All the successive ionisation energies of nitrogen are shown in the table.

Ionisation number	1	2	3	4	5	6	7
Ionisation energy / kJ mol^{-1}	1402	2856	4578	7475	9445	53267	64360

Explain the trend in the successive ionisation energies of nitrogen.

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(2 marks)